Multi-Agent Web Accessibility Improvement System

CS 568 Group 51

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Problem Statement and Motivation

Problem Statement:

Ensuring UI accessibility is challenging, as current tools offer limited, rule-based checks and lack AI-driven capabilities to automatically **identify and recommend fixes** for diverse accessibility issues

Motivation:

1. Accessibility is essential, not optional: Over 1 billion people globally experience some form of disability; many digital platforms remain hard to use or navigate

2. Existing tools fall short:

- Tools like AXE or Lighthouse are rule-based and miss nuanced context or personalized needs
- Developers lack smart assistants that **proactively suggest meaningful, user-centered fixes**

3. Opportunity for Human-AI Collaboration:

- Leverage AI not just to detect issues but to **simulate different user perspectives** (e.g., color blindness, motor impairments)
- Empower designers/developers to **empathize and act** using automated, intelligent insights

Inspiration and Need Finding

- **User-Centered Perspective**: Simulates diverse user experiences (visually impaired, motor-impaired, etc.) to inform accessible design

- **Augments Human Judgment**: AI agents don't replace developers—they support them with context-aware, explainable suggestions

- **Data-Driven Feedback**: Incorporates user study feedback to verify accessibility fixes—pushing HCI toward more participatory design processes

- **Intelligent Interaction Systems**:
  - Moves beyond static guidelines into dynamic, multi-agent systems
  - Facilitates natural language explanations and suggestions aligned with how humans reason about design issues

- **Evaluation via Real Users**: Validated through user studies to ensure usefulness, trustworthiness, and usability of the recommendations

Related Work

1. **Advancing Web Accessibility: Leveraging Generative AI for Enhanced WCAG Compliance** - Explores the use of generative AI to automate routine evaluations and offer advanced refactoring methodologies to enhance WCAG compliance

2. **From Cluttered to Clear: Improving the Web Accessibility Design for Screen Reader Users in E-commerce With Generative AI** - Provides a browser plugin that uses generative AI to restructure webpage content in real time, improving navigation for screen reader users on e-commerce platforms

3. **AI in Web Development**: **Enhancing Accessibility** - Discusses the incorporation of AI into web development to make it more accessible, including AI-driven accessibility audits and personalized user experiences

4. **WebUI**: **A Dataset for Enhancing Visual UI Understanding with Web Semantics** - Provides a large collection of rendered web pages with extracted metadata, aiding in visual UI understanding

Related Work

5. **Rico: A Mobile App Dataset for Building Data-Driven Design Applications** - Rico offers a dataset of mobile app interfaces to support data-driven design applications

6. **Evaluating Large Language Models in Generating Synthetic HCI Research Data** - This study evaluates the use of large language models to generate synthetic HCI research data, assessing the realism and utility of such data

7.**CoAuthor: Designing a Human-AI Collaborative Writing Dataset** - CoAuthor captures interactions between humans and AI during collaborative writing tasks, providing insights into AI's role in creative processes

RQ1: Can we generate an LLM-augmented dataset for accessibility fixes from web UI data?

How We Answered:

- **WebUI7k + AXE Violations**:
  - Started with WebUI7k, a dataset of web UI screenshots and code
  - Ran AXE tool to annotate existing **rule-based accessibility violations**
- **LLM-Augmented Fix Generation**:
  - Trained/fine-tuned multiple agents (Semantic, Contrast, Image Captioning) on this data
  - Each agent learned to detect a specific violation type and **generate context-aware, natural language fixes**
- **Created Fixes Dataset**:
  - Compiled a new dataset that includes:
    - Web page code/snippet
    - Detected violation (from AXE and LLM)
    - Fix recommended by the fixing agent

RQ2: Can AI agents recommend meaningful, user-centered fixes for accessibility issues?

How We Answered:

- **Agent-Specific Fix Recommendations:** Each fine-tuned agent (Semantic, Contrast, Image Captioning) independently analyzes violations and generates targeted fixes
- **Fixing Agent**: Aggregates outputs from all agents and recommends fixes
- **Fixes Dataset**: Created a new dataset pairing real-world websites, violations, and **agent-generated fixes**

Justification of Methodology:

- Used **multi-agent architecture** to simulate **different user personas** (color-blind, motor-impaired, visually impaired.)
- Fixes were not generic — they were:
  - Specific to **element-level** issues
  - Explained using **natural language** to improve developer understanding

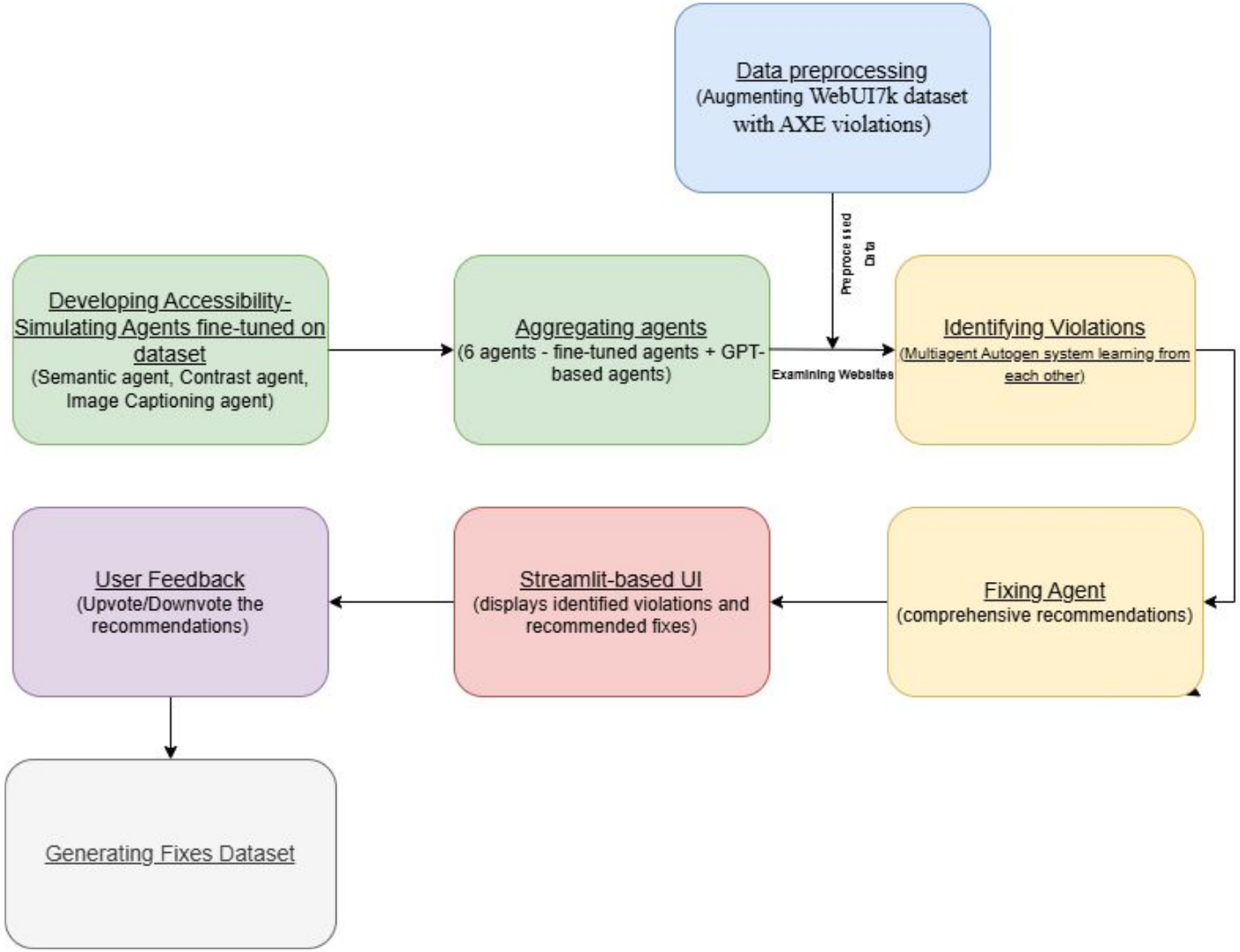
RQ3: Are the AI-generated fixes perceived as useful and actionable by users?

How We Answered:

- **User-Centric Evaluation**:
  - After generating fixes, we conducted a **user study** where participants reviewed and rated each fix
  - Users were shown the original UI snippet, the identified violation, and the agent-generated fix

- **Feedback Process**:
  - Users rated each fix on key dimensions:
    - **Usefulness**
    - **Clarity**
    - **Actionability**
  - Collected **quantitative scores** of upvotes & downvotes

Model Workflow



DEMO

Evaluation Study

- **Feedback Platform:** https://ai-assistant-for-accessibility.streamlit.app/

- **Scope of Study:**
  a. 11 real-world websites evaluated
  b. ≈ 4 accessibility issues and proposed fixes per site
  c. 14 unique violation categories tested

- **Participant Insights:**
  a. ~13 reviewers per website (total estimated reviewers: 142)
  b. 548 total votes submitted (upvotes + downvotes)
  c. Average reviewer agreement with fixes: **84.07%**

- **Link to Results:**

  https://docs.google.com/spreadsheets/d/1lijyxpdbrLFmPJQO8CTqlZzZaQ-6Rpa4TXA4SE7eWbQ/edit?usp=sharing

Evaluation Results

Flaws our agents already handle extremely well (>90 % reviewer agreement & ≥0.9 up-vote ratio)

- **Landmark – missing <main>**
- **Links – non-discernible text**
- **Links – non-discernible style**
- **Alt-text – inaccurate**
- **<html lang> missing**

Flaws that still need human review (<80 % agreement or many down-votes)

- **Missing <title> element**
- **High positive tabindex**
- **Heading hierarchy (H1 missing / level skips)**

Inter-dependent fix clusters

- **Alt-text quality** - *Missing alt ⇄ Inaccurate alt*
- **Landmarks** - *Missing <main> ⇄ Content outside landmarks ⇄ Duplicate landmarks*
- **Headings** - *H1 missing ⇄ Incorrect heading order*
- **Link affordance** - *Discernible text ⇄ Discernible style*

Independent fixes (votes show no correlation)

- **Colour-contrast adjustments – purely visual; orthogonal to structural fixes**
- **<html lang> attribute – language metadata only**
- **Positive tabindex changes – keyboard order, unrelated to landmarks/headings**

Future Scope

1. Fine-Tuning Agents with Human Feedback

- Retrain agents using our verified fixes dataset
- Improve accuracy and alignment with real user preferences

2. Expanded Disability Simulation

- Add agents for cognitive, hearing, and language-specific needs
- Broaden the range of simulated accessibility experiences

3. Explainable AI for Accessibility

- Provide natural language justifications for each fix
- Boost developer trust with transparent, interpretable suggestions

4. Real-Time Developer Tooling

- Launch a browser extension for live AI-powered accessibility auditing
- Empower designers with immediate, actionable feedback

References

1. Vemuri, V. P. K. (2024). *Advancing Web Accessibility: Leveraging Generative AI for Enhanced WCAG Compliance*. Journal of Artificial Intelligence & Cloud Computing, 3(2), 1–6. https://doi.org/10.47363/JAICC/2024(3)E261
2. Yu, Y., Ryskeldiev, B., Tsutsui, A., Gillingham, M., & Wang, Y. (2025). *From Cluttered to Clear: Improving the Web Accessibility Design for Screen Reader Users in E-commerce With Generative AI*. arXiv preprint arXiv:2502.18701. https://arxiv.org/abs/2502.18701
3. Madu, I., & Nzenwa, C. (2024). *AI in Web Development: Enhancing Accessibility*. SSRN. https://doi.org/10.2139/ssrn.4949516
4. Wu, J., Wang, S., Shen, S., Peng, Y.-H., Nichols, J., & Bigham, J. P. (2023). *WebUI: A Dataset for Enhancing Visual UI Understanding with Web Semantics*. arXiv preprint arXiv:2301.13280. https://arxiv.org/abs/2301.13280
5. Deka, B., Huang, Z., Franzen, C., Hibschman, J., Afergan, D., Li, Y., Nichols, J., & Kumar, R. (2017). *Rico: A Mobile App Dataset for Building Data-Driven Design Applications*. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology* (pp. 845–854). https://doi.org/10.1145/3126594.3126651
6. Hämäläinen, H., et al. (2023). *Evaluating Large Language Models in Generating Synthetic HCI Research Data: A Case Study*. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. https://doi.org/10.1145/3544548.3580688
7. Lee, N., et al. (2022). *CoAuthor: Designing a Human-AI Collaborative Writing Dataset*. arXiv preprint arXiv:2201.06796. https://arxiv.org/abs/2201.06796

Thank You!